Monetary Policy in Practice

CHAPTER 15 · PRINCIPLES OF MACROECONOMICS

Part 2: Policy in Practice, Money, Output, and the Long Run

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Roadmap

- Monetary policy in practice: inflation targeting, the output gap, the Taylor rule, the zero lower bound, and QE.
- Money, output, and the long run: short- vs long-run effects and monetary neutrality.
- Long-run changes: movements of money supply and money demand, then one checkpoint.

Inflation Targeting and the Output Gap

Indicator	Meaning
Inflation target	The inflation rate the central bank aims for over time. The Fed’s longer-run goal is 2% PCE inflation.
Output gap	How far real GDP is above or below potential GDP, as a percentage of potential GDP.

Output gap = 100 × (Y − Y*) / Y*

- If Y = 980 and Y* = 1,000, the output gap is −2%.
- Use inflation and output forecasts to guide policy. The Fed also seeks maximum employment.

Federal Reserve Board: inflation objective · Federal Reserve Board: policy rules

The Taylor Rule

r*: neutral real rate	π: inflation	π*: inflation target	g: output gap
2%	4%	2%	−2%

$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5g$$

- Start with $r^* + \pi = 2 + 4 = 6\%$.
- Inflation above target adds 1 point; the negative output gap subtracts 1 point.
- Suggested policy rate: 6%. The rule is a benchmark, not an automatic decision.

Federal Reserve Board: policy rules

The Zero Lower Bound

- When the nominal policy rate is near 0%, there is little room for further conventional rate cuts.
- Cash pays 0% interest. Deeply negative rates encourage people to switch to cash.
- A Taylor rule may suggest a rate below zero. Other tools, including QE, can provide support.

The effective lower bound can be slightly below zero.

Federal Reserve Board: policy limits

Quantitative Easing

- QE: the central bank buys large amounts of longer-term bonds, such as Treasuries and mortgage-backed securities.
- The purchases raise demand for those bonds. Their prices rise and their yields fall.
- Lower longer-term borrowing costs can support consumption, investment, and aggregate demand.
- QE can help when short-term rates are near zero. It does not guarantee more bank lending.

Federal Reserve Board: asset purchases and longer-term rates

Short- vs Long-Run Effects

	Short run	Long run
Wages and prices	Adjust slowly	Have fully adjusted
Real output	Can move above or below potential	Returns to potential
Price level	Begins to change	Reflects the change in money supply

- Monetary policy can change spending and real output in the short run.
- A one-time money-supply change does not permanently change productive capacity.

ECB: short-run effects and long-run neutrality

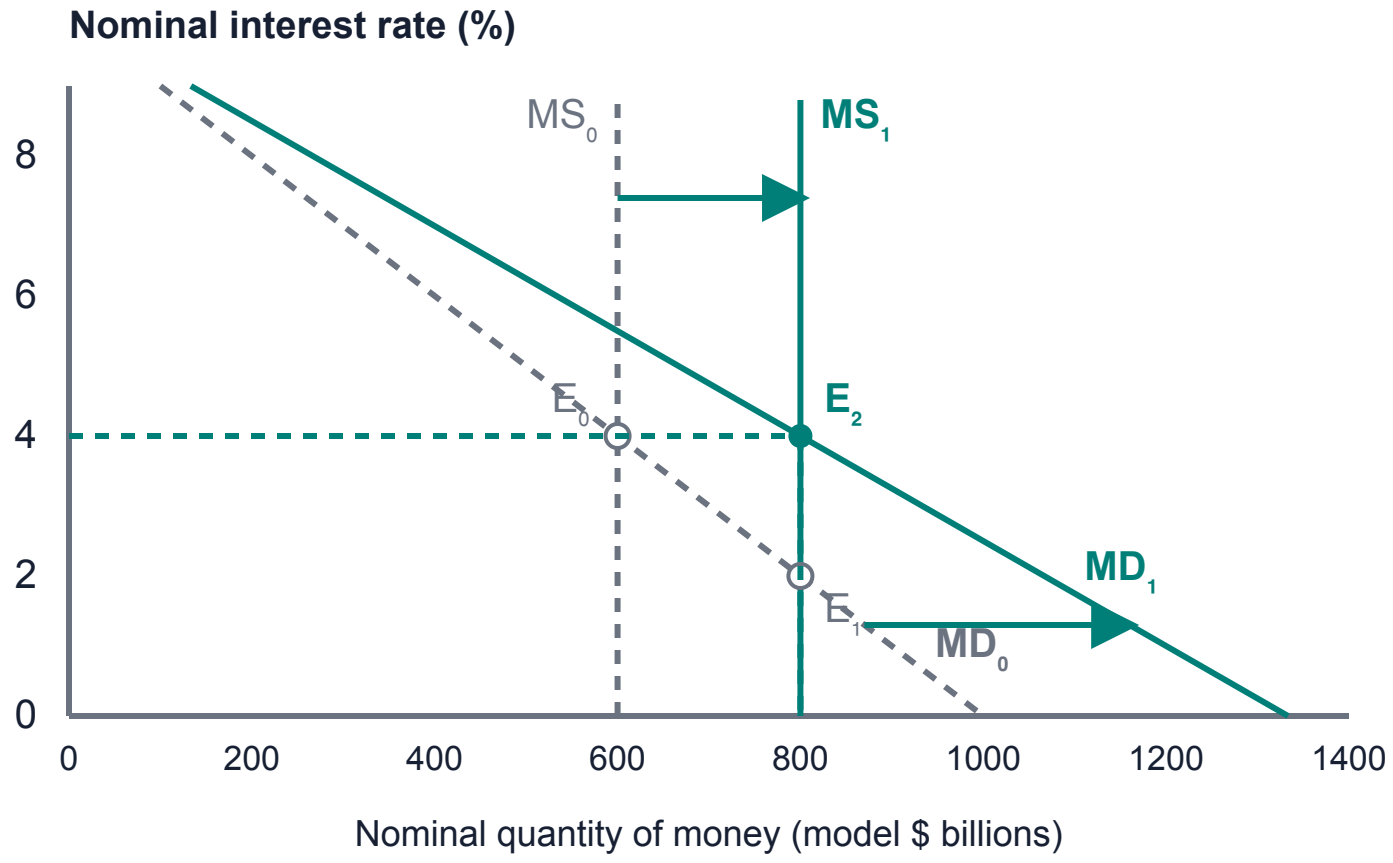
Monetary Neutrality

- Monetary neutrality: a change in money supply changes nominal variables, but not real output in the long run.
- If money supply rises by 10%, the long-run price level rises by 10%, with real output unchanged.
- More dollars do not create more workers, machines, or technology.

One-time change in money supply; full price adjustment; productive capacity unchanged.

ECB: short-run effects and long-run neutrality

Long-Run Changes: Monetary Expansion

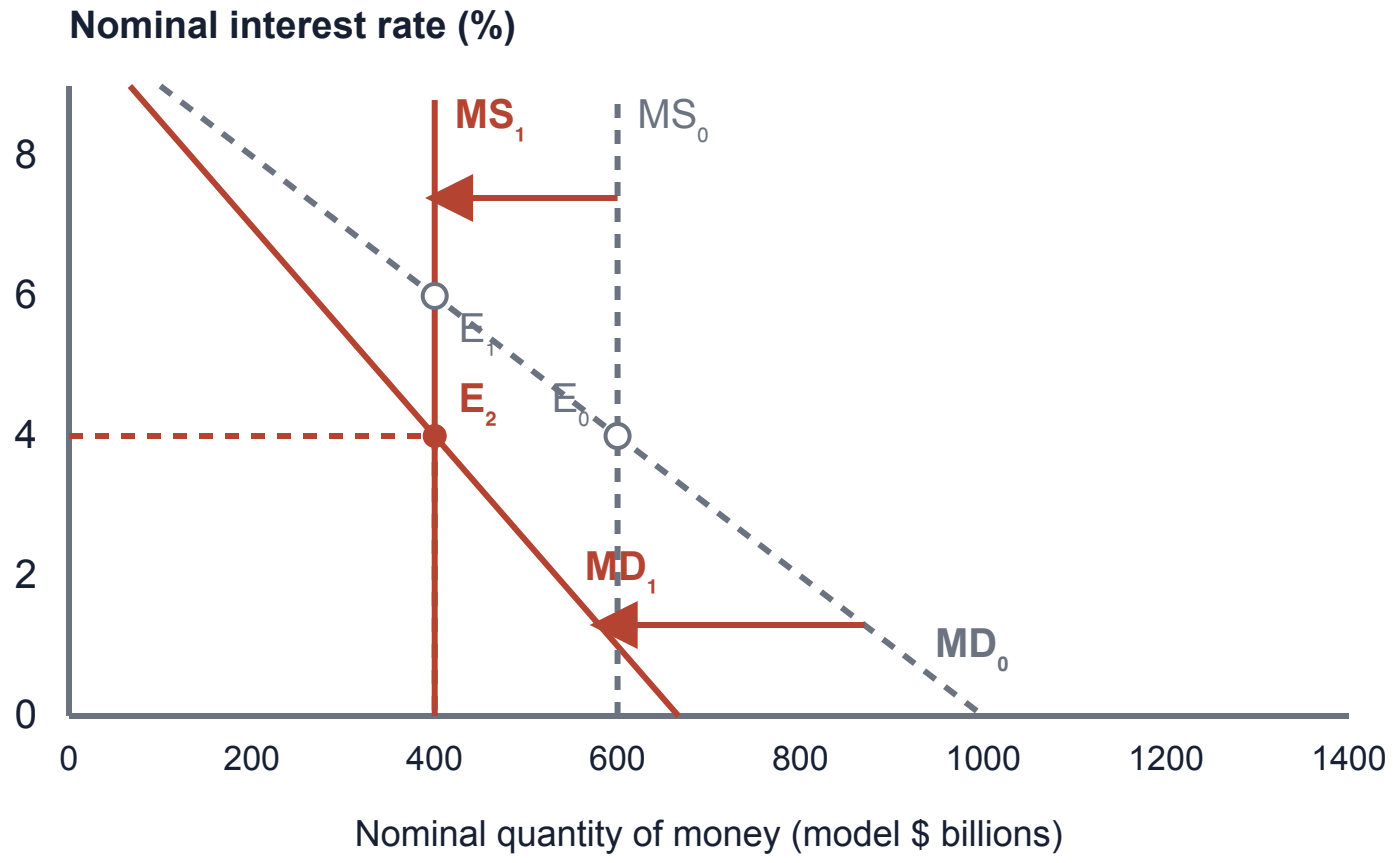


Long run: prices change; MS stays fixed.

1. Start: money supply is 600 and the interest rate is 4%.
2. MS shifts right to 800. With prices still fixed, the rate falls to 2%.
3. As prices rise, MD shifts right. In the long run, the rate returns to 4%; real output is unchanged.

ECB: short-run effects and long-run neutrality

Long-Run Changes: Monetary Contraction



Long run: prices change; MS stays fixed.

1. Start: money supply is 600 and the interest rate is 4%.
2. MS shifts left to 400. With prices still fixed, the rate rises to 6%.
3. As prices fall, MD shifts left. In the long run, the rate returns to 4%; real output is unchanged.

ECB: short-run effects and long-run neutrality

Monetary Neutrality Checkpoint

- After a one-time increase in money supply and full price adjustment, does real output remain above potential? Explain the change in money demand.

Explanation. No. Real output returns to potential. Higher prices raise nominal money demand; the long-run price level is higher.

ECB: short-run effects and long-run neutrality